

ATCD Assignment - 01

1. a) What is a string give an example
- b) Define and example for an alphabet

• A string is a finite sequence of symbols taken from an alphabet Σ

Ex: if $\Sigma = \{0, 1\}$ then examples of string are 0, 1, 01, 10, 11, 1101

Similarly if $\Sigma = \{a, b\}$ then formed strings are a, b, ab, ba, aab

- b) An alphabet is finite non empty set of symbols used to form strings

Ex: $\Sigma = \{0, 1\} \rightarrow$ binary alphabet

$\Sigma = \{a, b\} \rightarrow$ two letter alphabet

- Represented using the symbol ' Σ '
- These symbols are basic indivisible units
- Automata generally reads symbols directly instead of words

from the example 0 and 1, a and b are symbols whereas $\{a, b\}$ and $\{0, 1\}$ are alphabets.

- Alphabets are used like building blocks
- where as strings are combination of blocks.
- The number of symbols in an alphabet is known as cardinality of that alphabet.
- denoted using $|\Sigma|$

Ex: $\Sigma = \{a, b, c\}$.

$$|\Sigma| = 3.$$

- Every alphabet has a ϵ as a valid string.

Ex: $\Sigma = \{a, b\}$

valid strings = $\epsilon, a, b, aa, ab, ba, \dots$ etc.

2Q. Describe the structural representation of finite automata with a neat diagram.

A finite automaton (FA) is a mathematical model of computation used to recognize patterns generally known as language. This can be represented as 5-tuples.

$$FA = \{Q, \Sigma, \delta, q_0, F\}$$

Each structure can be explained as:

1. Set of States:

- Set of states is represented with Q .

where Q is a finite, non empty set of states. States represent different conditions / situation of automaton.

Ex: $Q = \{q_0, q_1, q_2\}$

2. Alphabet:

- Finite set of input symbols
- Automaton can only read one symbol at a time

Ex: $\Sigma = \{0, 1\}$

3. Transition function:

δ defines movement between states.

- It defines from current state on a given input where to go next?

Ex: $\delta(q_0, 0) = q_1$

4. Initial State:

The starting state of the automaton

- represented using a arrow ($\rightarrow$)
- only one initial state for every automaton.

5. Final State:

Final state in the automaton can be explained as set of states where the string is accepted.

Represented using double circles.

Ex. $F = \{q_2\}$

Diagrammatic Representation:

Consider a transition diagram as:



from the transition diagram

Transition diagram

$$Q = \{q_0, q_1, q_2\}$$

$$\Sigma = \{0, 1\}$$

$$\delta \Rightarrow \delta(q_0, 0) = q_1$$

$$q_0 = q_0 - \text{Initial}$$

$$F = q_2 - \text{final}$$

Transition table can be drawn as:

States	Inputs	
	0	1
$\rightarrow q_0$	q_1	q_0
q_1	q_2	ϕ
(q_2)	ϕ	q_2

Transition Table

Columns - represents input values

rows - transition from current state to next state.

from the table ϕ = dead state

which means the current state can't traverse to any path.

3. a) What is meant by Kleene closure
b) list any two applications of regular expressions

Kleene closure of a language L can be defined as the set of all possible strings formed from language by concatenating zero or more strings from L

let us represent this mathematically as.

$$L^* = \bigcup_{n=0}^{\infty} L^n$$

Ex: let $L = \{ab\}$.

$$L^0 = \{\epsilon\}$$

$$L^1 = \{ab\}$$

$$L^2 = \{abab\}$$

$$L^3 = \{ababab\}$$

$$L^* = L^0 \cup L^1 \cup L^2 \cup L^3 \dots L^n$$

$$L^* = \{\epsilon, ab, abab, ababab \dots\}$$

* Applications of regular expressions

1. Pattern matching in text processing: Regular expressions are widely used to improve the search and match patterns in text.

Ex: Regex = cat.

Matches = Blackcat, Cat123

- used in log analysis
- text editors.
- word processors

2. Finite automata construction: Every regular expression can be converted to NFA

- used to construct DFA and NFA for language recognition

4) Explain the relationship between finite automata and Regular expressions
Finite automata and Regular expressions are two different ways of representing the same class of language known as Regular languages.

• A language is regular if and only if it can be described by a regular expression and accepted by a finite automaton.

This explains

i) every regular expression has an equivalent finite automaton.

ii) every finite automaton has an equivalent regular expression.

i) Regular expression $\rightarrow$ Finite Automaton

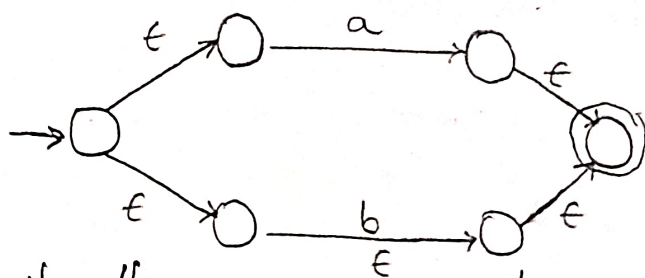
Expressions having regular languages can be converted from regular expression to finite automata using the state elimination reverse method known as state decomposition method.

• This method uses construction of NFA which can be further converted into DFA.

Ex: let us consider a regular expression and convert this expression into finite automata.

i) $(a+b)^* a \cdot b$

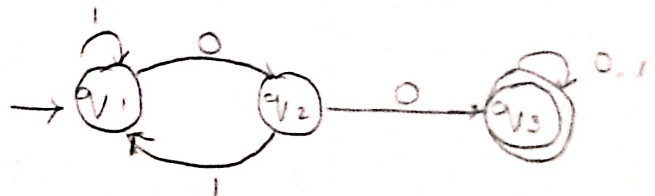
Step-1: let us initially represent $(a+b)$ this can be performed as



Here the union of a and b is performed.

Step-2: representing $(a+b)^*$

From step-1 the Kleene closure operation for concatenated input symbols of a & b can be obtained as



Step-1: from the transition diagram, for every node derive the product of input symbol and upcoming node only should be considered for the initial node ϵ should be added.

$$\therefore q_1 = q_1 \cdot 1 + q_2 \cdot 1 + \epsilon$$

$$q_2 = q_1 \cdot 0$$

$$q_3 = q_2 \cdot 0 + q_3 \cdot 0 + q_3 \cdot 1$$

Step-2: Substitute the values of q_1, q_2 in q_3 and solve the equations.

$$q_1 = q_1 \cdot 1 + q_1 \cdot 0 \cdot 1 + \epsilon$$

$$q_1 = q_1 (1 + 0 \cdot 1) + \epsilon$$

comparing with $R = Q + RP$ then $R = QP^*$

$$q_1 = \epsilon (1 + 0 \cdot 1)^* = (1 + 0 \cdot 1)^*$$

$$q_3 = q_2 \cdot 0 + q_3 \cdot 0 + q_3 \cdot 1$$

$$q_3 = q_1 \cdot 0 + q_3 (0 + 1)$$

$$q_3 = (1 + 0 \cdot 1)^* + q_3 (0 + 1)$$

$$R = Q + RP$$

$$R = QP^*$$

$$q_3 = (1 + 0 \cdot 1)^* 0 (0 + 1)^* \text{ is the final RE for FA.}$$

5. Construct an NFA that accepts all strings over $\{a, b\}$ end with abb

As per the question given that $\Sigma = \{a, b\}$ from which NFA accepts all strings ending with abb .

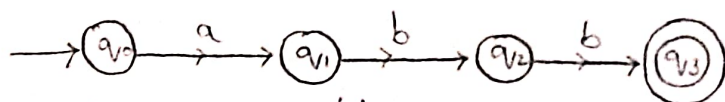
which means it accepts $abb, aabb, babb, aaabb$ etc.

for this NFA can be created as.

$$Q = \{q_0, q_1, q_2, q_3\}$$

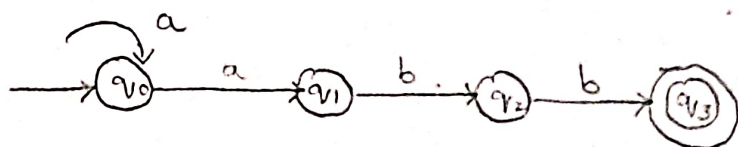
$$\Sigma = \{a, b\}$$

Step-1: construct a transition tree using Q as



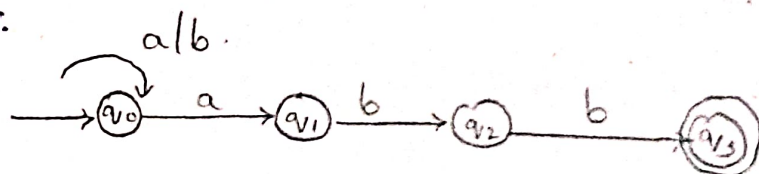
Now the given condition is verified as it follows abb but the formation of NFA is not done yet.

Step-2:



Self loop of input a for q_0 also not satisfies the condition of NFA.

Step-3:



NFA for $\{a, b\}$ ending with abb

Transition Table:

Current State	a	b
q_0	q_0, q_1	q_0
q_1	ϕ	q_2
q_2	ϕ	q_3
q_3	ϕ	ϕ